P510/2
PHYSICS
Paper 2
(Practical)
Sept. /Oct, 2026
3 hours

SET 4

Uganda Advanced Certificate of Education

PHYSICS

Paper 2

3 hours

INSTRUCTIONS TO CANDIDATES:

This examination paper consists of two compulsory items.

Use **blue** *or* **black** *ink. Any work done in pencil* **except** *graphs and diagrams will* **not** *be scored.*

Scores are awarded mainly for a clear record of the observations actually made, for their suitability, accuracy and for the use made of them.

Details on the question paper should not be repeated in the responses.

Graph paper is provided.

Silent non-programmable scientific calculators may be used

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Item 1.

At a construction site, an engineer observes that a crane uses a spring mechanism to support and control the movement of a heavy load. The spring is the main component of the mechanism, whose force constant is estimated to lie in the range **25.0 N m^{-1} to 45.0 N m^{-1}**. He observes that when heavier loads are attached, the spring stretches more than when lighter loads are attached.

Interested in establishing the cause of this behaviour, the engineer consults a technician who advises that the properties of the spring should be verified using a scientific investigation. The technician also has a damaged spring balance at his workshop and wishes to replace its spring with one having similar properties to the original spring. You are provided with a spring believed to have similar properties to the one used in the crane mechanism.

Task:

Investigate the properties of the spring provided to explain to the engineer why the spring stretches by different amounts for different loads and determine whether it is suitable for use in repairing the spring balance.

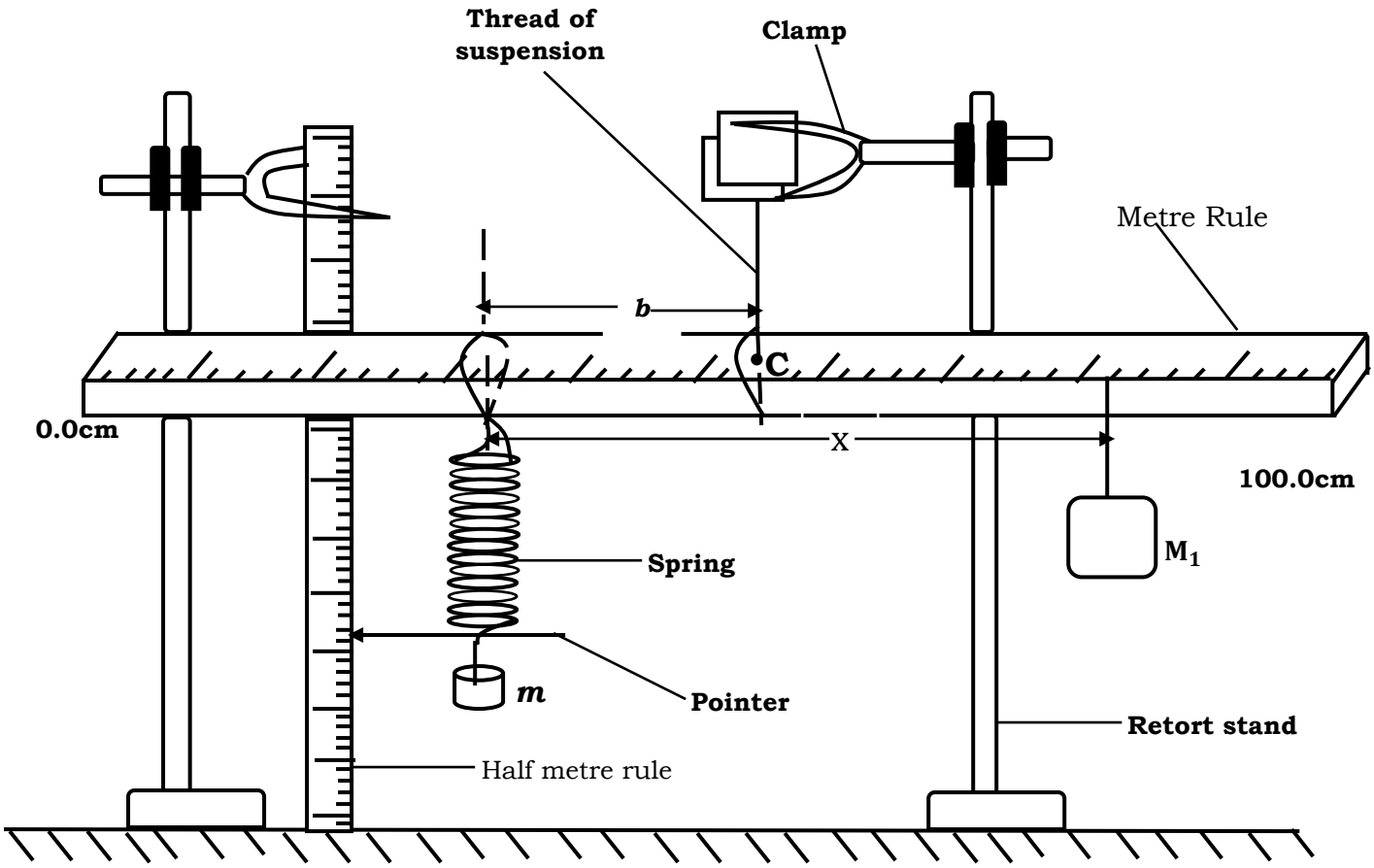
METHOD I

a) Suspend the metre rule from the retort stand clamp with its scale facing upwards using a piece of thread.

b) Suspend the spring at the 40.0cm mark of the metre rule.

c) Adjust the position of the loop of the thread until the metre rule balances horizontally as shown in ***figure 1*.**

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d) Locate the point C at which the metre rule balances horizontally from the 0.0cm mark.
e) Locate the point C at which the metre rule balances horizontally from the 0.0cm mark.
f) Find the distance, b, of the spring from point C.
g) Read and record the initial position, $\mathbf{p_o}$ of the pointer from the half metre rule (in metres)
h) Suspend a mass $m = 0.050kg$ from the lower free end of the spring, and a mass $M_1 = 0.200kg$ at the 50.0cm mark of the metre rule.
i) Adjust the position of M_1 until the metre rule balances horizontally again.
j) Read and record the new position, **p**, of the pointer (in metres)
k) Calculate the extension, **e**, of the spring.
l) Measure and record the distance x
m) Repeat procedures (g) to (k) for values of $m = 0.100,\ 0.150,\ 0.200,\ 0.250$ amd $0.300kg$
n) Record your results in a suitable table including values of $y = x - b$ and $\frac{y}{b}$.
o) Plot a graph of $\frac{y}{b}$ against e

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p) Find the slope, S, of the graph.

q) Calculate the values of K_1 from the expression $K_1 = \frac{Sg}{5}$

Where $\boldsymbol{g = 9.81ms^{-2}}$

METHOD II

a) Clamp the spring provided using the two pieces of wood provided and attach a pointer to the free end of the spring, making sure that the spring is vertical and the pointer is horizontal as shown in ***figure 2***.

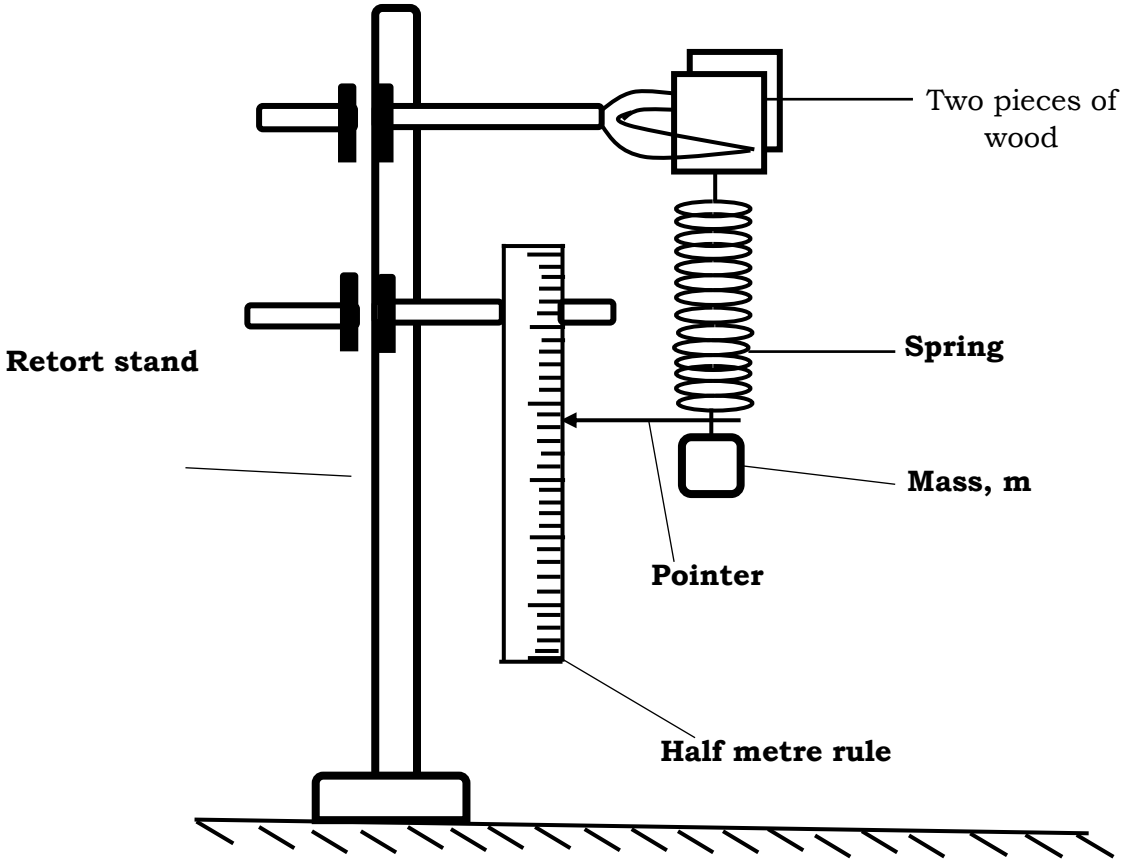


Figure 2

b) Read and record the initial position, y_o of the pointer on the half metre rule.

c) Suspend a mass, $m_1 = 0.200$kg from the lower free end of the spring.

d) Read and record the new position, y_1 of the pointer.

e) Find the extension $x_1 = y_1 - y_o$ (in metres) produced in the spring.

f) Replace the mass m_1 with $m_2 = 0.300$kg.

g) Read and record the position of the pointer, y_2 on the half metre rule

h) Find the extension $x_2 = y_2 - y_o$ (in metres).

i) Find the value of from the expression $K_2 = \frac{(m_2 - m_1)g}{x_2 - x_1}$.

Where $g = 9.81ms^{-2}$

j) Calculate the constant, K of the spring from the expression:

$$2K = K_1 + K_2$$

Item 2.

A health centre in a remote community uses a solar heating system to warm water for cleaning medical equipment. The system uses a concave mirror to concentrate sunlight onto a small container. However, the workers are uncertain whether the available mirror is suitable for the system.

The mirrors available at the health centre have unknown focal lengths. From the manufacturer's specifications, a suitable mirror should have a radius of curvature in the range 36.0 cm to 40.0 cm to concentrate sunlight effectively.

A laboratory technician advised the workers to use two methods to establish the suitability of the mirror:

The workers, however, do not know the appropriate experimental procedures to follow. They have therefore approached you for assistance and provided one of the concave mirrors for investigation.

Task:

Investigate to determine the suitability of the concave mirror provided for use in the solar heating system.

End